

AI-BASED RESUME DETECTION AND ANALYSIS SYSTEM



A MINI-PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

It's becoming more and more important for job seekers to be able to expertly tailor their resumes for every application to further their careers in the fast-paced industry. The purpose of this project is to change the way resume analysis and job profile matching are done by introducing the use of Natural Language Processing (NLP) techniques, mainly the Large Language Models (LLMs). The conventional methods of resume data extraction and job matching have used only NLP, which is the reason why they have not been able to achieve the best accuracy and diversity. The use of LLMs to improve the quality and quantity of the insights that are extracted from the resumes, thus, the LLMs will be able to match the candidate profiles with the job requirements more precisely. Our approach consists of several important aspects. Firstly, we use NLP to extract the necessary data from the resumes, which will guarantee a full data capture. Thus, by applying resume matching, we improve the match of job profiles and candidate can easily find jobs whose skills match with their profiles. In addition, our application has a feature of a resume builder that helps to make resumes suitable for job roles using the information extracted and the analysis of the personality. Through the resolution of the existing methodologies gap, our project establishes a new level of resume analysis and job matching, and thus enables the candidates to find their way to their preferred job.

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LIST OF ABBREVIATIONS

NLP	-	Natural Language Processing
HTTP	-	Hypertext Transfer Protocol
LLM	-	Latent Language Models
NLTK	-	Natural Language Toolkit
XML	-	Extensible Markup Language
API	-	Application Programming Interface
DOC	-	Document
NER	-	Named Entity Recognition
UI	-	User Interface
BERT	-	Bidirectional Encoder Representations from Transformers
GPT	-	Generative Pre-trained Transformer

CHAPTER 1

INTRODUCTION

1.1 Introduction

Particularly in the dynamic career development sector, the innovative integration of Natural Language Processing (NLP) into the Resume Analyzer signifies a paradigm shift in the conventional practice of resume design for job applications. This Resume Analyzer explains the complex relationships between natural language processing algorithms and how they may revolutionize the hiring process by suggesting strategic certifications and identifying relevant job roles.

Using natural language processing (NLP) and open-source LLM, the Resume Analyzer reimagines the resume customization process and situates it at the intersection of cutting-edge technology and professional advancement. The tool predicts a candidate's intended career job using advanced natural language processing (NLP) techniques, going beyond resume analysis. It also looks at what NLP does while evaluating resumes to gather qualifications, credentials, experience, and skills. It is based on the NLP breakthrough, an advanced technological development paradigm, making it suitable for the Resume Analyzer.

Resume Analyzer employs natural language processing and machine learning techniques, which provide a detailed analysis of the textual data in resumes. The system shall analyze information related to the applicant's background data to find out compatible job positions. The system accurately arrives at the right employment function for an individual based on detailed scrutiny of job descriptions and the incorporation of industry terminologies. Now with the addition of machine learning,

The Resume Analyzer is much more than a statistical tool. A prediction machine that the company claims could alter how prospective employees think about their future career paths. The Resume Analyzer is also forecasted as it promotes specific qualifications that are in line with the required job description. This system takes into consideration several factors relevant to the candidate's competencies available in the markets and the general state of the industry. In addition to task prediction, the objective gives users useful information and directs them toward Certifications that might enhance their standing in the field. By positioning the Resume Analyzer as a holistic career development tool, this proactive approach fosters a culture of continuous learning and skill development. Our approach consists of several important aspects. First, we use NLP for the extraction of necessary information from the resume, which will ensure that all data is captured. Therefore, by applying the resume matching, we improve the match of the job profiles, and candidate can find jobs whose skills match their profiles easily. Besides, our application includes a feature of a resume builder that helps to make resumes suitable for job roles using the information extracted and analysis of personality.

1.2 Problem Statement

Job seekers face a difficult barrier that might take two different shapes. To begin with, conventional resume modification processes find it difficult to satisfy the demands of different job applications. These techniques frequently lack precision and inventiveness. Because of the shortcomings of these traditional approaches, professionals are unable to adequately showcase their skills, which results in missed opportunities and a decreased likelihood of landing the job they want. The problem is made worse by the lack of a proactive tool that recommends certificates based on job opportunities.

In the absence of strategic leadership, professionals are discouraged from making deliberate attempts to enhance their professional profiles and keep abreast of industry trends. A creative resume analyzer that goes above and beyond conventional procedures is much needed to handle these two problems. This application requires the use of contemporary Natural Language Processing (NLP) techniques to assess job obligations and determine the best place to work. In addition to serving as an assessment tool, the Resume Analyzer ought to be a proactive collaborator for experts, providing astute suggestions for certificates that correspond with the designated job description. Because it enables job searchers to proactively expand their skill sets and manage the intricacies of the modern labor market, this proactive component is essential. The intended Resume Analyzer turns into a strategic ally for job seekers navigating the modern labor market, rather than just a useful tool. The application addresses the first aspect of the issue by guaranteeing a thorough evaluation of resume content through the use of sophisticated natural language processing (NLP) techniques. The algorithm takes care of the second dimension at the Same time, by accurately determining the ideal employment position and recommending credentials that improve the professional profile. Having access to a proactive and all-inclusive solution like these aids in career development and helps job seekers better handle the demands of the modern labor market.

The creation of a proactive and intelligent Resume Analyzer that integrates state-of-the-art natural language processing algorithms, predictive features, and customized certification suggestions is ultimately required to address this issue. In doing so, this instrument turns into a priceless tool that aids job seekers in effectively and mindfully navigating the complexities of the employment market.

1.3 Objectives

- Implement NLP to extract data like personal information, skills, education, and work experience from the resume.
- Compare multiple prompts to get the optimal result from the Large Language Model.
- To verify whether the application process offers candidates instant feedback while they upload their resumes.

1.4 Significance and Motivation of the Project Work

The Resume Analyzer project is notable since its objective is to improve the job application process. These days, a tool like this one is needed to accurately assess some of a candidate's skills because many employment marketplaces are very competitive. This project aims to assist people in properly showcasing themselves through a polished narrative to improve their chances of landing desired positions. The challenges people have when attempting to prepare resumes served as the inspiration for the notion. Many people find it challenging to identify critical abilities and align them with market demands. This is when the Resume Analyzer comes in handy. By providing tailored guidance and recommendations, it reduces this strain and increases the user's sense of comfort and empowerment during the job application process. The second objective of the project is to advance an individual's professional growth. The Resume Analyzer helps people advance their careers and stay competitive by recommending relevant qualifications and adapting to changing market conditions. In the end, it'll be like having a helpful professional success travel companion, with practical advice that directly affects job seekers' chances of getting hired.

CHAPTER 2

LITERATURE SURVEY

2.1 Literature Survey

S. No.	Author and Year	Tools/Techniques/Dataset	Methodology	Results	Limitations
1	Gan, C., Zhang, Q., & Mori, T. (2024a, January 16).	Natural Language Processing, Open-source LLMs, Conditional Random Fields, Vectorization Models	Focuses on leveraging LLM agents for various stages of the automated resume screening process, from sentence classification to decision-making, to enhance efficiency and effectiveness in candidate selection.	These results collectively showcase the effectiveness, efficiency, and performance improvements achieved through the implementation of LLM agents in the automated resume screening framework.	The limitations of the automated resume screening framework based on LLM agents include data compromise risk, varying model capabilities, and reliance on fine-tuning, resource-intensive human feedback requirement, performance variability, and the need for customization to generalize across industries.

2	Jagwani V, Meghani S, Pai K, Dhage S. Published July 28, 2023.	SpaCy, Latent Dirichlet Allocation model	Uses SpaCy's Named Entity Recognition to extract relevant entities and an LDA model to rate resumes by assigning topic probabilities to each entity.	The proposed method successfully extracts relevant entities and assigns topic probabilities to each entity using LDA.	Doesn't provide diversity over multiple career domains
3	Bhor, S., Gupta, V., Nair, V., Shinde, H. and Kulkarni, M.S., 2021. Int J Res Eng Sci (IJRES), 9(6), pp.01-06.	Conditional Random Fields (CRF), Convolutional Neural Network (CNN), Optical Character Recognition (OCR)	CNN for segmentation and extraction of information from a resume. Named Entity Recognition (NER) is used to understand domain-specific jargon and words	Extract details and statistics from resumes and rank them based on company preferences and requirements. The output JSON contains various extracted information like personal, educational details, etc.	The accuracy of the NLP technique used, as the effectiveness of the resume parser heavily relies on it. The performance may also be influenced by the quality and format of the resume being parsed.

4	Daryani, C., Chhabra, G.S., Patel, H., Chhabra, I.K. and Patel, R., 2020. ETHICS AND INFORMATION TECHNOLOGY [Internet] . VOLKS ON PRESS, pp.99-103.	Natural Language Processing Toolkit, Term-Frequency - Inverse Document Frequency , Cosine-Similarity matrix	Uses Natural Language Processing to extract relevant information and a cosine-similarity matrix to match a resume with the job description.	Presents an automated resume screening system that simplifies the recruitment process for organizations.	Does not address the limitations of using NLP in accurately capturing the candidate's qualifications
5	Green, N., Liu, M. and Murphy, D., 2020. Information Systems Education Journal, 18(3), pp.28-37.	NLTK, Term-Frequency-Inverse Document Frequency, Cosine-Similarity	Uses Natural Language Processing to extract relevant information and a cosine-similarity matrix to match a resume with the job description.	Presents a tool that utilizes NLP that analyze resumes and produce visual reports that highlight areas for improvement.	Does not address the biases of using NLP techniques

(2.1 Literature Survey for Resume Analyzer Using NLP)

2.2 Key Gaps in the Literature

The literature on automated resume screening and parsing has a few significant gaps and areas for possible improvement, based on the summaries that have been offered. The following are some observations:

A Basic Discussion of Ethical Issues: While Natural Language Processing (NLP) is mentioned by Chirag Daryania, Gurneet Singh Chhabra, Harsh Patel, Indrajeet Kaur Chhabra, and Ruchi Patel [1], the ethical issues surrounding automated resume screening are not discussed. Crucial elements that must be considered and covered in the literature are ethical considerations, such as potential discrimination and bias in NLP models.

Insufficient Diversity in Career Fields: According to the second paper, it doesn't offer diversity across a range of career fields. To create inclusive hiring procedures, diversity is crucial. Scholars ought to investigate strategies for augmenting the generalizability of their approaches across diverse businesses and occupational domains.

Lack of Restrictions Talk: Many publications don't specifically address the restrictions or difficulties encountered while putting their resume parsing algorithms into use. It is vital to acknowledge and deliberate upon the constraints to comprehend the pragmatic suitability and possible drawbacks of the suggested techniques.

Comparative analysis and evaluation metrics: While the articles describe the tools, approaches, and methodologies used, they do not go into great length about the assessment criteria that are used to gauge how well the suggested systems operate.

A more comprehensive understanding of the efficacy of the suggested models may be obtained by conducting comparative evaluations with current practices or benchmarks.

NLP Technique Biases: Certain publications don't discuss the biases that come with using NLP approaches. Unfair results might arise from biases in language models, particularly in the recruitment process. An essential component of study in this field should be debating methods to reduce biases and enhance fairness.

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CHAPTER 3

SYSTEM DEVELOPMENT

3.1 Requirements

Python: The Basis for Readability and All-Inclusive Libraries

Python is a popular, advanced, and versatile programming language. Its vast libraries and ease of reading have contributed to its global appeal. It is quite convenient for many operations like data analysis, web-building, etc.

- **Readability:** The readability and relative ease with which to write and maintain coding for developers is highly emphasized in the syntax of Python, which plays an important role herein. This feature will be significant in large codebases and collaborative projects.
- **Large Library:** The enormous library and framework ecosystem within Python is what makes it possible for the tool to fulfill multiple demands. These include TensorFlow, PyTorch, Django, Flask, NumPy, Pandas, and many others. Development is made easier as these libraries are accessible while giving programmers strong tools.

Text Analysis's Untapped Potential: The Natural Language Toolkit (NLTK)

A special Python package for use in NLP known as NLTK, which stands for Natural Language ToolKit. The program provides numerous functionalities, including sentiment analysis, stemming, and tokenization, to name some.

- **Text Analysis capabilities:** NLTK stands out as one of the top NLP tools because it has numerous capabilities for text analysis.

- For example, developers can employ features such as word tokenization, which breaks down text into words, and stemming, which reduces words back to their root form, among others.
- **Natural Language Processing:** Enhanced computational abilities of a computer using NLTK will be possible, which makes it possible for programmers to create computers that can understand, read, and produce text that sounds as if a human wrote it.

spaCy: Robust Advanced Natural Language Understanding

This software is known for being effective, having pre-trained models, and supporting various tasks such as named entity recognition, tokenization, and part-of-speech tagging.

- **Efficiency and Speed:** SpaCy is commonly recognized to be fast in handling NLP tasks.

It has been optimized for production use, which will run a large amount of text data using minimal resources.

- **Text Extraction:** Textual data can be handled and studied inside PDF files thanks to PDFMiner's ability to extract text content from PDF documents.
- **Structured Data Extraction:** PDFMiner can extract structured data from PDFs in addition to text. This is especially helpful for PDF documents that contain data in tabular or sorted formats.
 - **Resume Parsing:** PyResparser was developed specifically to parse resumes and retrieve vital data, including contacts, work history, education, and talents. The amount of manual labor required to review resumes is greatly decreased by this automation.

- **Structured Data Output:** Integrating the collected data into databases or application tracking systems is made easy by the library's structured data output.

WebParser: Web scraping for better job matching

BeautifulSoup4 is a flexible Python module made to make online scraping easier and more accessible for developers to extract data from XML and HTML sources.

- **Parse HTML and XML:** Developers can parse web content and extract structured data using BeautifulSoup4's intuitive parsing methods, making it easy to navigate and extract information from complex document structures.
- **Navigate Document Structure:** The library provides tools for traversing the hierarchical structure of HTML and XML documents, enabling developers to locate and extract specific elements or data points efficiently.
- **Extract Information:** BeautifulSoup4 offers powerful methods for extracting information from web pages, including `find()`, `find_all()`, and `select()`, allowing developers to retrieve relevant data with ease.

Humantic API: Personality insights of the candidate according to the resume

The Humantic API is designed to analyze text data using advanced artificial intelligence and natural language processing (NLP) techniques, with a specific focus on sentiment analysis and emotion recognition. Here's what the Humantic API does:

- **Sentiment Analysis:** The API evaluates text data to determine the sentiment expressed within it. It can classify text as positive, negative, or neutral based on the language used and the overall tone of the content.

- **Emotion Recognition:** In addition to sentiment analysis, the Humantic API can recognize specific emotions conveyed in the text. It can identify emotions such as happiness, sadness, anger, fear, and others, providing insights into the emotional context of the text.
- **Customizable Parameters:** The API offers customizable parameters that allow users to fine-tune the analysis according to their specific requirements. Users can adjust parameters such as language settings, sentiment thresholds, and emotion categories to optimize results for different use cases and applications.

3.2 Project Design and Architecture

A major development in resume analysis and optimisation is represented by the Resume Analysis Application using Latent Language Models (LLM). The software provides job searchers wishing to improve their resumes and increase their chances of success in the cutthroat job market with a complete solution that combines advanced natural language processing techniques with an intuitive UI.

One of the main advantages of the programme is its intuitive and smooth user interface, which has been meticulously created. Users are led through the resume analysis process by an aesthetically pleasing and tidy interface as soon as they launch the programme. Resumé uploading in PDF, DOC, and DOCX formats is made easy for users by the interface's straightforward drag-and-drop file upload capability. All along the way, users are helped and made to comprehend every stage of the analysis process by the clear explanations and tooltips.

Following an upload of a resume, the programme starts the preparation phase, which involves normalising and cleaning the text to guarantee correctness and consistency. Ensuring uniform resume content and analysis readiness depends on this preprocessing stage. To make more analysis easier, formatting errors and irrelevant material are eliminated, and the text is standardised in design.

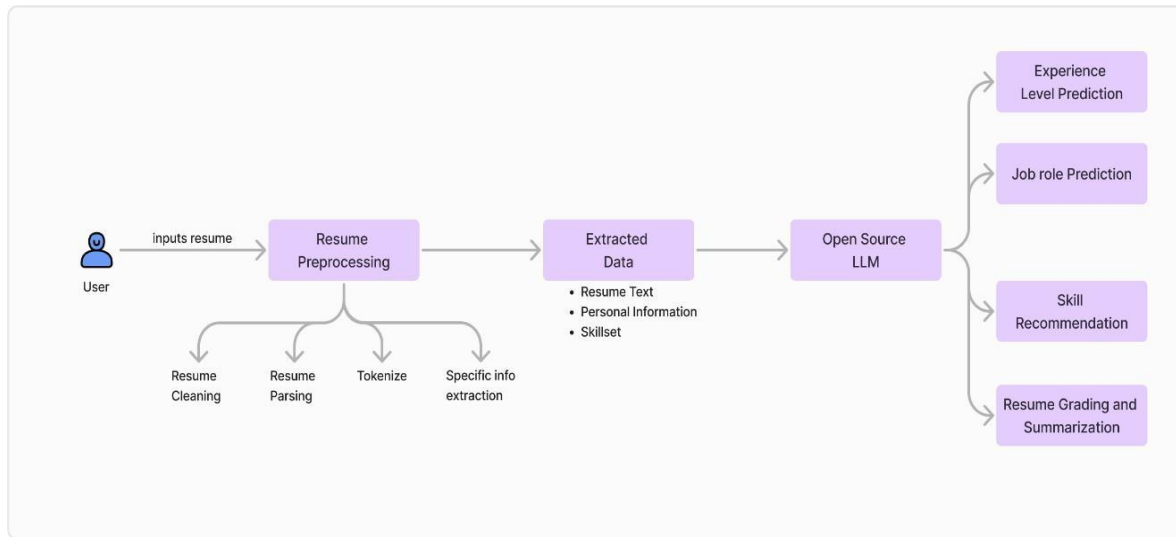


Fig.3.1: system flow chart

Modern natural language processing models trained on enormous volumes of text data, known as BERT or GPT, provide the basis of the analysis engine of the programme. These models let the programme decipher the semantic background of the resume text and get useful information from it. Employing sophisticated methods such as entity identification, semantic similarity, and context-aware language understanding, the software can extract important data from the resume, including work experience, education, abilities, certifications, and accomplishments.

The application's analysis goes beyond simple keyword matching to give customers a thorough understanding of the advantages, disadvantages, and areas for development of their resumes. Clients get customised suggestions based on their particular skill set, professional objectives, and market needs.

Among these suggestions are ways to highlight important qualifications and experiences, streamline resume material, and make layout changes. To help users fill up skill gaps and progress in their professions, the software also compares extracted abilities and experiences with job criteria and industry trends and offers suggestions for pertinent courses, certifications, and training programmes. The analytical algorithms and recommendations are always being updated as people use the programme and offer input. Through this iterative approach, users may optimise their resumes and succeed in their job hunt by knowing that the application is always current and responsive to changing user needs and market trends. All things considered, the Resume Analysis Application with Latent Language Models (LLM) is a major development in resume analysis technology that gives job searchers practical information and customised suggestions for professional growth.

3.3 Data Preparation

Automated systems are now essential for quickly processing and evaluating candidate data in the modern job market, when hiring managers and recruiters are deluged with resumes. The resume cleansing and resume parsing are two essential parts of these systems. We'll examine the nuances of both processes in this thorough overview, looking at their goals, important duties, underlying methods, and overall integration inside automated systems.

Approaches: - Regular expressions for pattern matching, string matching algorithms for phrase recognition, and context-aware noise reduction using natural language processing (NLP) approaches are some of the techniques used during resume cleaning. Moreover, machine learning models can be applied to more difficult jobs like standardizing and formatting.

The results of the resume cleaning procedure serve as the basis for the automated system's later phases.

The information input into the resume processing step is consistent and free of extraneous noise when the data is cleaned and standardized, which creates the conditions for the precise extraction of crucial information.

Resume Parsing:

The next step in automated resume analysis is called resume parsing, and it involves methodically organizing and structuring the cleaned and standardized data for simple analysis and retrieval. The main goal is to gather precise information in an organized manner regarding a candidate's experiences, abilities, and credentials. During the hiring process, insights are generated, and judgments are made based on this structured data.

1. Entity Extraction: - Extracting important entities from the cleansed data is one of the main jobs in resume parsing. This entails locating and retrieving personal data (name, contact information), as well as data on educational background, professional experience, certificates, and other pertinent information. For this objective, Named Entity Recognition (NER) models are frequently used.

2. Contextual Understanding: - To comprehend the contextual relationships among various entities in the resume, algorithms are used. For example, contextual comprehension is necessary to discern between a common term and a specific ability. This guarantees that the system understands the text's complex meaning correctly.

3. Normalization: - Transforming extracted data into a standard format is the process of normalization. This includes maintaining a unified representation of data overall, standardizing job titles, and making sure date formats are consistent. Normalization is essential to removing discrepancies that could occur from variations in the information that candidates choose to include on their resumes.

4.Database Integration: - For archiving and retrieval, the parsed data is incorporated into a database. This database facilitates effective searching and analysis by acting as a central store for candidate data.

3.4Key Challenges

Creative Resumes and Unstructured Data: For automated systems, managing creative resume forms and unstructured data presents a major difficulty. Certain candidates use eye-catching but non-traditional layouts, like infographics, graphics, or nonlinear structures. It takes sophisticated natural language processing (NLP) methods and machine learning models that can accurately adjust to a variety of resume formats to extract pertinent information from such resumes. Finding the right balance in data extraction between precision and flexibility is still a difficult task.

Ambiguities and Nuances in Language: Resume analysis systems are constantly faced with the subtleties and ambiguities found in natural language. Frequently, candidates utilize terminologies that are context-dependent, industry-specific, or abbreviated in ways that differ throughout industries. Comprehending the minute variations in linguistic subtleties is essential to precise entity identification and contextual comprehension. To improve the system's language Understanding capabilities, this task calls for constant training on a variety of datasets, semantic analysis, and NLP algorithm improvement.

Changing Industry Demands and Skill Sets: Resume analysis systems have challenges due to the dynamic nature of skill requirements and the quick evolution of industries. The changing nature of work roles and the emergence of new technology may make traditional keyword-based techniques obsolete.

Systems must be flexible, constantly upgrading their knowledge of vocabulary and expertise unique to a certain business. It becomes essential to integrate with real-time job market data and industry professional feedback methods to stay up to date with evolving trends and guarantee that the system's recommendations are still applicable.

Data security and privacy concerns: Strong data security measures are necessary since resumes contain sensitive personal information, which causes privacy concerns. Resumes contain personal information that should be protected from unwanted access, including contact details and occasionally even sensitive identifiers. Addressing privacy concerns and building user trust in automated resume analysis systems requires implementing encryption techniques, adhering to data protection legislation, and adopting safe data storage practices.

CHAPTER 4

TESTING

4.1 Testing Strategy

4.1.1 Unit Test:

The NLP Model Unit Test, which verifies the precision and efficacy of the Natural Language Processing (NLP) model, is a crucial component of the Resume Analyzer project's overall testing approach. This test seeks to ascertain whether the NLP model accurately extracts pertinent data from a range of resumes. In this context, it entails presenting the model with many templates that depict diverse conventions, styles, or forms adopted by industries. In the unit test, the model determines if it can process input items in text format correctly, detect key elements, as well as generate useful information about work experience, academic qualification, and other related details. Throughput assessment on such a microscopic level makes sure that the interpretation of the NLP model matches the complex subtleties of actual practice. The successful execution of the NLP Model Unit Test demonstrates the validity of the model and acts as the basis for the competence of the Resume Analyzer to analyze different resumes correctly.

4.1.2 Integration Test:

To test the reliability of the Resume Analyzer system, end-to-end testing should be carried out to ensure that interfaces match perfectly. Integration is necessary for the NLP model, certification recommendation, and job prediction modules to function as a cohesive unit. This entails ensuring that data flows seamlessly between modules and that consistency and correctness are always upheld. Next, end-to-end testing is used to evaluate the system's functionality in detail.

This testing begins with the resume input and concludes with the outputs of job prediction and certification recommendation. This test simulates real-world environments to confirm that the related parts work well together and produce correct results. A well-functioning system that assesses resumes, predicts job openings, and integrates and recommends relevant certificates is one that integrates and tests well from start to finish. Regular validation and refinement of these procedures improve the Resume Analyzer project's overall dependability and efficacy.

4.1.3 User-Interface Test:

The success of the project depends on efficient input validation, which ensures that the Resume Analyzer system handles a range of resume formats and input styles with grace. This means setting up dependable procedures to interpret and preprocess different resume formats while considering variations in document types, layouts, and content configurations. Its capacity to manage this variability makes the system more flexible and easier to use, allowing it to easily accept the diverse range of resumes that are submitted for review. But output validation is equally crucial because it verifies that the system displays certification and job projections accurately.

4.2 Test Cases and Outcomes

In this section, test cases are used to assess the system, and the anticipated results are covered in this section.

Test Case no. 1

Test Case No.	1
Objective:	To test whether the user can upload the file from the file explorer.

Test Data	Expected Result	Actual Result
Click the Upload Button and upload the file	Open File Explorer and upload the file.	Open File Explorer and upload the file.

Test Result	The above result indicates that the admin can click the upload button and upload files.
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Test Case no. 2

Test Case No.	2
Objective:	To test whether the necessary information from the resume is parsed.

Test Data	Expected Result	Actual Result
Upload the resume	Show name, email, mobile number, technical skills, and soft skills.	Show name, email, mobile number, technical skills, and soft skills.

Test Result	The above result indicates that the necessary information from the resume has been parsed.
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CHAPTER 5

RESULTS AND EVALUATIONS

5.1 Results

Resume Analyzer

The results obtained from the resume analyzer show that the system can extract resume data from the PDF of any format (alignment, text style, fonts, etc.), given as input. It can successfully extract personal information of the user, like name, contact list, and the number of pages in the resume. In addition to the above results, it can highlight the skill set of the user, predict their targeted job role, and recommend skills that can be added to their resume to increase their chances of landing their targeted job. It also provides an elaborate grading of all the components in the resume, i.e., education, skills, etc., and overall grading of the resume. In addition, it provides recommendations to make required changes in the resume to make it better.

Recommendation

- Implement an AI-powered system to streamline resume screening and evaluation.
- Use Natural Language Processing (NLP) to extract relevant information from resumes.
- Train models to identify key sections: education, experience, skills, certifications, etc.
- Integrate Named Entity Recognition (NER) for accurate entity extraction (names, companies, dates).
- Apply text classification models to evaluate resume relevance against job descriptions.

CHAPTER 6

CONCLUSIONS AND FUTURE ENHANCEMENT

In conclusion, the Resume Analyzer using NLP is a smart, innovative, and interactive tool that can automatically analyze resumes. Natural Language Processing techniques enable the tool to predict the profession the user is interested in. Furthermore, its ability to recommend skills and certifications, using a hardcoded dataset that can be added to the user's resume, assists individuals in meeting industry requirements. By providing personalized inputs and recommendations, this tool not only guides users toward better opportunities but also serves as a valuable career development partner.

The incorporation of intelligent solutions, such as a scoring system, resume data storage in a database, and an administrative dashboard, is how hiring processes are expected to operate in the future. This proposed system centralizes and organizes information while introducing a data-driven approach to resume evaluation. One essential element is the Administrator Dashboard, which offers a centralized area for in-depth analysis and tactical decision-making. Recruiters can gain useful insights into the makeup and progress of the applicant pool with its features, which include Diversity Metrics, Distribution of Skills and Keywords, Application Progress Overview, and Resume makeup Analysis. The database-based resume data storage solution guarantees a seamless and well-organized hiring process concurrently. An organized schema, scalability, data security, and interface with existing systems are important factors to consider when designing a database analysis and retrieval.

APPENDICES

APPENDIX I: SOURCE CODE

FRONTEND CODE

```
<!DOCTYPE html>

<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
  <title>Contact Us - Resume ATS Checker</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-
alpha1/dist/css/bootstrap.min.css" rel="stylesheet">
  <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-
icons@1.10.3/font/bootstrap-icons.css">
  <link rel="stylesheet" href="{ { url_for('static',
filename='css/styles.css') } }">
</head>
<body>
  <!-- Navigation -->
  <nav class="navbar navbar-expand-lg navbar-dark bg-dark">
    <div class="container">
      <a class="navbar-brand" href="/">Resume ATS Checker</a>
      <button class="navbar-toggler" type="button" data-bs-
toggle="collapse" data-bs-target="#navbarNav">
        <span class="navbar-toggler-icon"></span>
      </button>
      <div class="collapse navbar-collapse" id="navbarNav">
```

```

        <ul class="navbar-nav me-auto">
            <li class="nav-item">
                <a class="nav-link" href="/">Home</a>
            </li>
            <li class="nav-item">
                <a class="nav-link" href="/about">About</a>
            </li>
            <li class="nav-item">
                <a class="nav-link active" href="/contact">Contact</a>
            </li>
        </ul>
    </div>
</div>
</nav>
<!-- Contact Header -->
<div class="contact-header">
    <div class="container text-center">
        <h1>Contact Us</h1>
        <p class="lead">We're here to help with any questions or
feedback you may have</p>
    </div>
</div>
<div>
    <div class="col-lg-5">
        <div class="row g-4">
            <div class="col-md-12">
                <div class="card contact-card h-100">
                    <div class="card-body text-center">
                        <i class="bi bi-envelope contact-icon"></i>
                        <h4>Email Us</h4>

```

```

        <p>support@atsresumechecker.com</p>
        <p>info@atsresumechecker.com</p>
    </div>
</div>
</div>
<div class="col-md-12">
    <div class="card contact-card h-100">
        <div class="card-body text-center">
            <i class="bi bi-telephone contact-icon"></i>
            <h4>Call Us</h4>
            <p>+91 123456789</p>
            <p>Monday to Friday, 9 am to 5 pm EST</p>
        </div>
    </div>
</div>
</div>
</div>
</div>
</div>
<div class="py-4 mt-5">
    <div class="container text-center">
        <p>© 2023 Resume ATS Checker. All rights reserved.</p>
    </div>
</div>
</footer>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-alpha1/dist/js/bootstrap.bundle.min.js"></script>
<script src="{ { url_for('static', filename='js/theme.js') } }"></script>
</body>
</html>

```

BACKEND CODE

```
import tkinter as tk
from tkinter import filedialog, messagebox, scrolledtext
from customtkinter import CTk, CTkButton, CTkLabel, CTkTextbox, CTkEntry,
CTkFrame, CTkScrollableFrame, set_appearance_mode, set_default_color_theme
import PyPDF2
import docx
import re
from collections import Counter
import spacy
import os
import threading
from fpdf import FPDF

# Set appearance mode and color theme
set_appearance_mode("System") # Modes: "System" (standard), "Dark", "Light"
set_default_color_theme("blue") # Themes: "blue" (standard), "green", "dark-blue"

# Initialize spaCy with error handling
try:
    Nlp = spacy.load("en_core_web_sm")
except OSError:
    print("Downloading spaCy model...")
    os.system("python -m spacy download en_core_web_sm")
    nlp = spacy.load("en_core_web_sm")

# CustomTkinter App
class ATSanalyzerApp(CTk):
    def __init__(self):
        super().__init__()
        self.title("ATS Resume Analyzer")
        self.geometry("800x650")
        self.minsize(700, 600)
        # Mainframe
        self.main_frame = CTkFrame(self)
        self.main_frame.pack(fill=tk.BOTH, expand=True, padx=20, pady=20)
        # Header
        self.header_label = CTkLabel(self.main_frame, text="ATS Resume Analyzer",
font=("Arial", 20, "bold"))
        self.header_label.pack(pady=10)
        # Input frame (left side)
        self.input_frame = CTkFrame(self.main_frame)
```

```

self.input_frame.pack(fill=tk.BOTH, expand=True, padx=10, pady=10)
# Resume upload section
self.resume_frame = CTkFrame(self.input_frame)
self.resume_frame.pack(fill=tk.X, padx=10, pady=10)
self.resume_label = CTkLabel(self.resume_frame, text="Resume:")
self.resume_label.pack(side=tk.LEFT, padx=5)
self.resume_path = CTkEntry(self.resume_frame, width=300)
self.resume_path.pack(side=tk.LEFT, fill=tk.X, expand=True, padx=5)
self.upload_button = CTkButton(self.resume_frame, text="Upload Resume",
command=self.upload_resume)
self.upload_button.pack(side=tk.RIGHT, padx=5)
# Job Description section
self.job_desc_frame = CTkFrame(self.input_frame)
self.job_desc_frame.pack(fill=tk.BOTH, expand=True, padx=10, pady=10)
self.job_desc_label = CTkLabel(self.job_desc_frame, text="Job Description:")
self.job_desc_label.pack(anchor=tk.W, padx=5, pady=2)
self.job_desc_text = CTkTextbox(self.job_desc_frame, height=150)
self.job_desc_text.pack(fill=tk.BOTH, expand=True, padx=5, pady=5)
# Keywords section
self.keywords_frame = CTkFrame(self.input_frame)
self.keywords_frame.pack(fill=tk.X, padx=10, pady=10)
self.keywords_label = CTkLabel(self.keywords_frame, text="Additional
Keywords (comma separated):")
self.keywords_label.pack(anchor=tk.W, padx=5, pady=2)
self.keywords_entry = CTkEntry(self.keywords_frame)
self.keywords_entry.pack(fill=tk.X, padx=5, pady=5)
self.keywords_entry.insert(0, "Python, Java, SQL, Machine Learning, Data
Analysis, Teamwork, Communication")
# Analyze the button
self.analyze_button = CTkButton(self.input_frame, text="Analyze Resume",
command=self.start_analysis)
self.analyze_button.pack(pady=10)
# Get file path
file_path = self.resume_path.get().strip()
If not file_path:
    messagebox.showerror("Error", "Please upload a resume file first.")
    return
def _process_resume(self, file_path):
    try:
        # Extract text from resume
        self.resume_text = self.extract_text(file_path)

        # Get job description and additional keywords

```

```

    job_desc = self.job_desc_text.get("1.0", tk.END).strip()
    additional_keywords = self.keywords_entry.get().strip()
    # Update UI with results
    .update_results(score, recommendations)
except Exception as e:
    self.status_var.set(f"Error: {str(e)}")
    messagebox.showerror("Error", f"Failed to process the resume: {str(e)}")
    Try:
        Text += page.extract_text() + "\n"
    Except Exception as e:
        text += f"[Error extracting page: {str(e)}]\n"
    If not text.strip():
        messagebox.showwarning("Warning", "The PDF may be scanned or
image-based.
    Else:
        raise ValueError("Unsupported file format")
    return text
except Exception as e:
    raise Exception(f"Error extracting text: {str(e)}")
if additional_keywords:
    user_keywords = [k.strip() for k in additional_keywords.split(",") if k.strip()]
else:
    all_keywords = user_keywords
# Create case-insensitive pattern for keyword matching
text_lower = text.lower()
keywords_pattern = r'\b(' + '|'.join(re.escape(kw.lower()) for kw in all_keywords) +
r')\b'
# Check for contact information (10 points)
email_pattern = r'[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}'
phone_pattern = r'(\+?\d{1,3}[\s-])?(\d{3})?[\s-]?d{3}[\s-]?d{4}'
linkedin_pattern = r'linkedin\.com/in/[a-zA-Z0-9_-]+'
# Check for section headers (max 30 points)
sections = ["experience", "education", "skills", "projects", "summary", "objective"]
# Check resume length (10 points)
word_count = len(text.split())
if 300 <= word_count <= 600: # Changed from 700 to 600
    length_score = 10
elif 200 <= word_count < 300 or 600 < word_count <= 700: # Changed upper
bounds
    length_score = 5
else:
    length_score = 0
if keyword_score < 30:

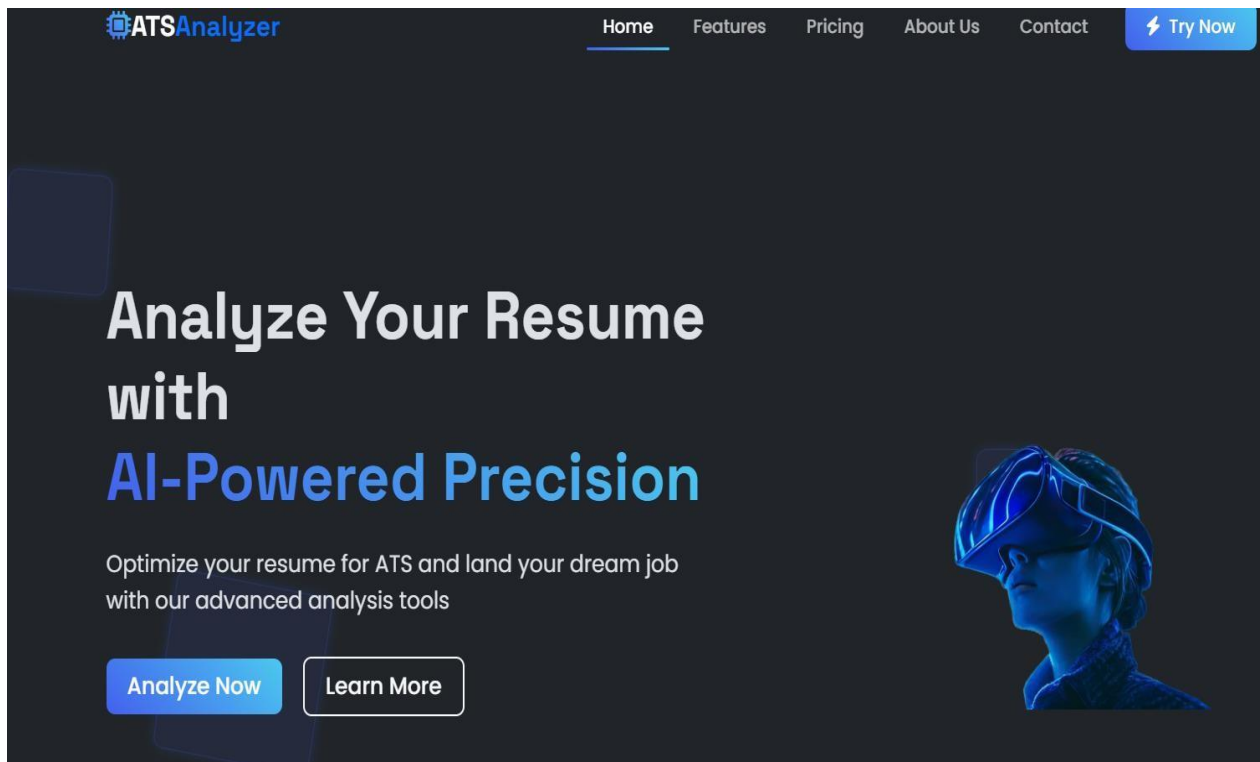
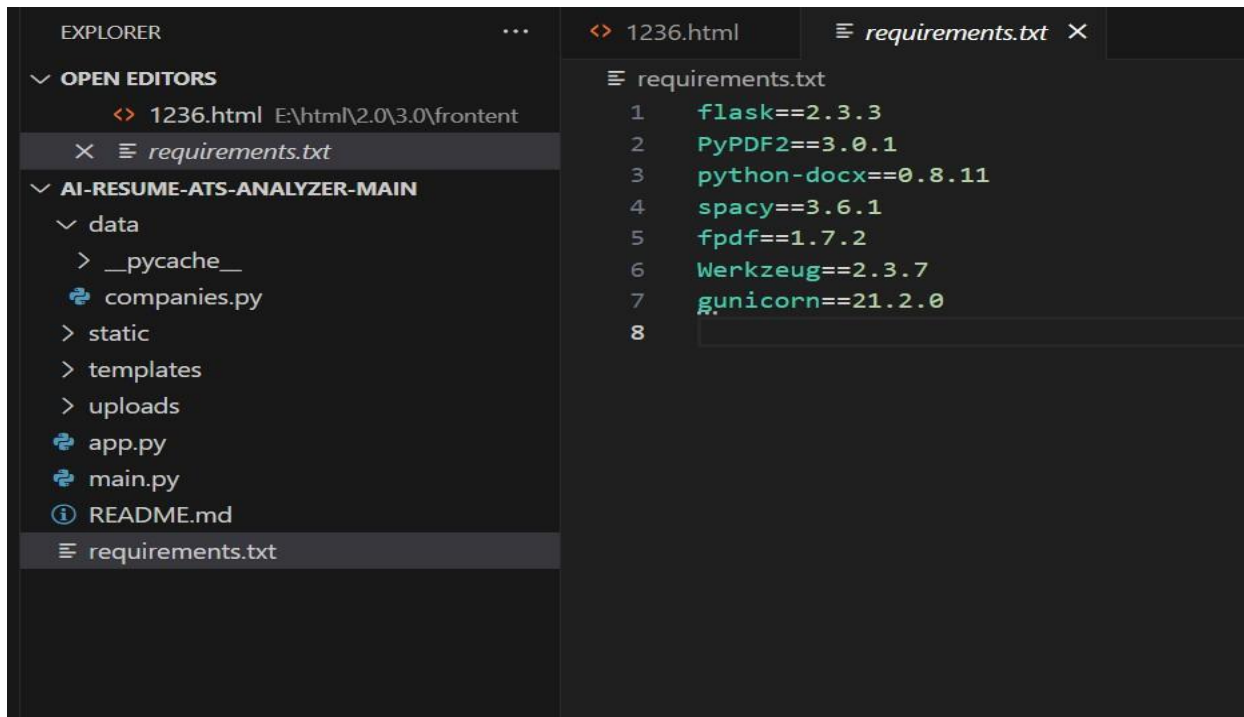
```

```

        missing_keywords = [kw for kw in all_keywords if kw.lower() not in text_lower]
        if missing_keywords:
            recommendations += f"1. Add these missing keywords: {'',
'.join(missing_keywords[:10])}"
            if len(missing_keywords) > 10:
                recommendations += f" and {len(missing_keywords) - 10} more"
            recommendations += "\n\n"
        else:
            recommendations += "1. Keyword usage is good!\n\n"
# Contact info recommendations
if contact_info_score < 10:
    recommendations += "2. Improve contact information:\n"
    if not has_email:
        recommendations += " - Add your email address\n"
    if not has_phone:
        recommendations += " - Add your phone number\n"
    if not has_linkedin:
        recommendations += " - Add your LinkedIn profile\n"
if section_score < 30:
    missing_sections = [section for section in sections if section not in
found_sections]
    if total_score < 50:
        recommendations += "Significant improvements are needed."
    elif total_score < 75:
        recommendations += "Some improvements would help your resume stand out."
    else:
        recommendations += "Your resume is well-optimized for ATS systems!"
try:
    if file_path.endswith(".pdf"):
        self.export_to_pdf(file_path)
    elif file_path.endswith(".txt"):
        self.export_to_txt(file_path)
    messagebox.showinfo("Export Successful", f"Results exported to {file_path}")
except Exception as e:
    messagebox.showerror("Export Failed", f"Failed to export: {str(e)}")
# Run the app
if __name__ == "__main__":
    try:
        app = ATSAnalyzerApp()
        app.mainloop()
    except Exception as e:
        messagebox.showerror("Application Error", f"An error occurred: {str(e)}")

```


APPENDIX II: SCREENSHOT





NAME : MOSES.S
DEGREE : Diploma
BRANCH : Computer Science
COLLEGE : CSI Polytechnic College
MAIL : mosesmoses1953@gmail.com
MOBILE : 9361195039/9976102170

CARRER OBJECTIVE

Intent to build a career with leading corporate with committed and dedicated people, and to Excel my skills day by day which help me to explore myself fully and realize my potential. Willing to work with team proactively and as a key player in challenging and creative Environment.

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